UNIVERSITY PHYSICS REFERENCE HANDBOOK SERIES

VOLUME 8 • PURE ENGLISH MASTER EDITION

Waves & Sound

A comprehensive, mathematically rigorous synthesis of mechanical wave propagation, the 1D classical wave equation, acoustic energetics and decibel scaling, Doppler kinematics, standing resonant modes in strings and air columns, and acoustic beats with rapid hand-calculation fast-tracks.

CORE THEORETICAL & APPLIED DOMAINS

- 1D Classical Wave Equation & d'Alembert Solutions
- Longitudinal Sound in Gases & Adiabatic Bulk Modulus
- Sound Level & Logarithmic Decibel (dB) Scale
- Standing Waves, Boundary Inversion & Nodes
- Acoustic Beats, Envelope Modulation & Tuning
- Transverse Waves on Strings & Speed Scaling
- Wave Energy Transmission, Power & Intensity
- Master Acoustic Doppler Formulation & Echo Shifts
- Harmonic Series in Open and Stopped Organ Pipes
- 8 Hand-Calculation Fast-Tracks & 4 Worked Problems

Synthesized From: University Physics Lecture Courseware (*Wave and Sound*) & R. Shankar, *Fundamentals of Physics: Mechanics, Relativity, and Thermodynamics* (Yale University Press, Chapters 18 & 19).
Publication Standard: Pure English Technical Monograph • Vector SVG Schematics • Complete SI Derivations.

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Part I: Foundations of Wave Motion & The 1D Wave Equation

Chapter 1: Nature and Mathematical Representation of Waves

A **wave** is a self-sustaining disturbance that propagates through space and time, transferring momentum and energy from one location to another without transporting gross matter. Waves are broadly categorized into two fundamental classes based on their physical transmission mechanism:

- **Mechanical Waves:** Disturbances that require a deformable, elastic material medium (such as a string, water, solid rock, or air). The medium possesses both *inertia* (mass density) and *elastic restoring forces* (tension, elasticity, or pressure gradients) that allow adjacent elements to transmit disturbances. Examples include seismic P and S waves, acoustic sound waves, and string vibrations.
- **Electromagnetic Waves:** Coupled, oscillating electric and magnetic fields that propagate through the electromagnetic vacuum at the speed of light $c \approx 3 \times 10^8$ m/s, requiring no physical material carrier whatsoever.

1.1 Transverse vs. Longitudinal Wave Modes

The distinction between transverse and longitudinal motion depends upon the geometric relationship between the local particle displacement vector **s** and the wave propagation velocity vector **v**:

- **Transverse Waves:** Medium particles oscillate *strictly perpendicular* to the direction of wave travel (**s** $\perp$ **v**). Examples: transverse vibrations on a guitar string, electromagnetic waves. Transverse mechanical waves require shear rigidity and can only propagate through solids or along tensioned 1D boundaries (strings/membranes); they cannot propagate through the bulk of ideal fluids.
- **Longitudinal Waves:** Medium particles oscillate *strictly parallel* to the direction of wave travel (**s** $\parallel$ **v**). Examples: acoustic sound waves in fluids, compression springs. Longitudinal waves propagate as alternating regions of high density and pressure (**compressions** or condensations) and low density and pressure (**rarefactions**). All physical media (solids, liquids, and gases) possess bulk compressive elasticity, allowing longitudinal waves to travel through all states of matter.

1.2 Mathematical Description of a Harmonic Traveling Wave

Consider a continuous, sinusoidal (harmonic) wave propagating along the x-axis. The transverse displacement $y(x, t)$ of a medium element located at coordinate x at time t is given by the standard harmonic wave function:

General Harmonic Wave Function: $$y(x, t) = A \sin(kx - \omega t + \varphi) \quad [\text{Propagating in } +x \text{ direction}]$$

If the wave propagates in the negative x direction ($-x$), the sign inside the argument becomes positive: $y(x, t) = A \sin(kx + \omega t + \varphi)$. The constitutive parameters are rigorously defined as:

Parameter	Symbol & Definition	SI Unit	Physical Significance
Amplitude	A	m	Maximum transverse displacement from equilibrium ($y = 0$).
Wavelength	λ	m	Spatial repeat distance between consecutive crests or identical phase points.
Period	$T = 1 / f$	s	Temporal duration required for one complete cycle at a fixed position.
Frequency	$f = 1 / T$	Hz (s^{-1})	Number of complete oscillation cycles passing a fixed coordinate per second.

Parameter	Symbol & Definition	SI Unit	Physical Significance
Angular Frequency	$\omega = 2\pi f = 2\pi / T$	rad/s	Rate of phase advance in radians per unit time.
Wave Number	$k = 2\pi / \lambda$	rad/m	Spatial frequency: phase advance in radians per unit distance.
Phase Velocity	$v = \lambda f = \omega / k$	m/s	Speed at which a point of constant phase (crest/trough) propagates through space.
Phase Constant	φ	rad	Initial phase offset determined by initial conditions at $x = 0$, $t = 0$.

1.3 Particle Kinematics vs. Wave Propagation Velocity

A fundamental source of confusion in wave mechanics is conflating the **wave propagation speed** v with the **transverse particle velocity** v_y. The wave speed v describes the motion of the geometric disturbance and energy envelope through the medium, which is constant in a uniform medium. In stark contrast, individual particles do not travel along x; they execute local Simple Harmonic Motion (SHM) parallel to the y-axis:

Transverse Particle Velocity:
$$v_y(x, t) = \frac{\partial y}{\partial t} = -\omega A \cos(kx - \omega t + \varphi)$$

Transverse Particle Acceleration:
$$a_y(x, t) = \frac{\partial^2 y}{\partial t^2} = -\omega^2 A \sin(kx - \omega t + \varphi) = -\omega^2 y(x, t)$$

The maximum particle speed and acceleration attainable during simple harmonic oscillation are:

$$v_{y,\,max} = \omega A = 2\pi f A, \qquad a_{y,\,max} = \omega^2 A = 4\pi^2 f^2 A$$

Notice that $v_y = 0$ when the particle is at an extreme displacement (crest or trough, $y = \pm A$), where transverse restoring acceleration $|a_y|$ reaches its absolute maximum $\omega^2 A$. Conversely, when the particle crosses the equilibrium axis ($y = 0$), particle speed reaches $|v_y| = \omega A$ while instantaneous acceleration vanishes.

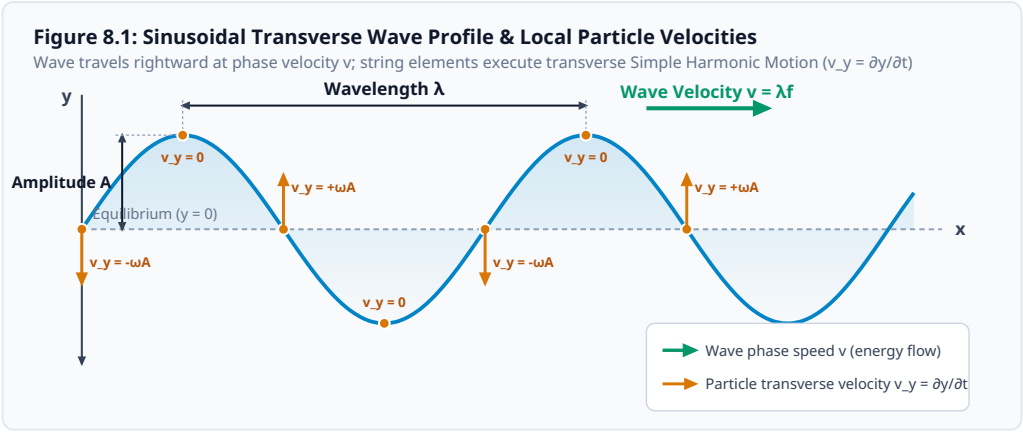


Figure 8.1: Instantaneous spatial profile of a rightward-traveling harmonic wave showing wavelength λ, amplitude A, phase velocity v, and local particle velocity vectors v_y.

Chapter 2: Derivation of the 1D Classical Mechanical Wave Equation

The classical wave equation is the master partial differential equation governing all small-amplitude, non-dispersive wave phenomena. We present a rigorous first-principles derivation by applying Newton's Second Law to a differential string element.

▶ First-Principles Derivation from Newton's Second Law

Consider a perfectly flexible, uniform string under constant equilibrium tension T, having a linear mass density $\mu = dm / dx$. When a transverse wave passes, an infinitesimal segment between coordinates x and $x + dx$ is displaced transversely by $y(x, t)$.

Under the small-amplitude approximation, the slope of the string at any point is very small: $|\partial y/\partial x| \ll 1$. Consequently, the angle θ between the string segment and the horizontal x-axis satisfies:

$$\sin\theta \approx \tan\theta = \frac{\partial y}{\partial x}, \qquad \cos\theta \approx 1$$

The horizontal components of tension at both ends balance exactly: $T\cos(\theta + d\theta) - T\cos(\theta) \approx 0$. Along the vertical y-axis, the net transverse restoring force dF_y is:

$$dF_y = T\sin(\theta + d\theta) - T\sin\theta \approx T\left[\left.\frac{\partial y}{\partial x}\right|_{x+dx} - \left.\frac{\partial y}{\partial x}\right|_{x}\right] = T\,\frac{\partial^2 y}{\partial x^2}\,dx$$

By Newton's Second Law, this net vertical force must equal the mass of the segment $dm = \mu\,dx$ multiplied by its transverse acceleration $a_y = \partial^2 y / \partial t^2$:

$$T\,\frac{\partial^2 y}{\partial x^2}\,dx = (\mu\,dx)\,\frac{\partial^2 y}{\partial t^2}$$

Dividing both sides by $T\,dx$ yields the celebrated **1D Classical Wave Equation**:

$$\frac{\partial^2 y}{\partial x^2} = \frac{\mu}{T}\,\frac{\partial^2 y}{\partial t^2} = \frac{1}{v^2}\,\frac{\partial^2 y}{\partial t^2}$$

Comparing coefficients reveals the wave speed v directly in terms of medium elasticity and inertia:

$$v = \sqrt{\frac{T}{\mu}}$$

2.1 The General d'Alembert Solution

The 1D linear wave equation is satisfied not only by sinusoidal waveforms, but by *any twice-differentiable arbitrary function* of the compound variables $(x - vt)$ or $(x + vt)$. This is known as **d'Alembert's Solution**:

d'Alembert's General Wave Solution: $\quad y(x, t) = f(x - vt) + g(x + vt)$

Here, $f(x - vt)$ represents an undistorted wave pulse of arbitrary profile traveling in the positive x direction at constant velocity $+v$, while $g(x + vt)$ represents a pulse traveling in the negative x direction at velocity $-v$.

Chain Rule Proof: Let $u = x - vt$. By the multivariable chain rule:

$\partial y/\partial x = f'(u) \cdot (\partial u/\partial x) = f'(u), \qquad \partial^2 y/\partial x^2 = f''(u)$

$\partial y/\partial t = f'(u) \cdot (\partial u/\partial t) = -v\, f'(u), \qquad \partial^2 y/\partial t^2 = (-v)^2\, f''(u) = v^2\, f''(u)$

Substituting these into the wave equation gives $f''(u) = (1/v^2)[v^2\, f''(u)] = f''(u)$, which is an exact identity for any arbitrary waveform f.

Part II: Wave Propagation Speeds in Diverse Media

Chapter 3: Mechanical Wave Speeds in Solids, Fluids, and Gases

Every mechanical wave speed is dictated by the universal ratio of an **elastic property** (restoring stress per strain) to an **inertial property** (mass density):

$$\mathrm{v} = \sqrt{\frac{\text{Elastic Property (Restoring Force)}}{\text{Inertial Property (Resistance to Motion)}}}$$

Medium & Wave Type	Governing Formula	Elastic Factor	Inertial Factor
Stretched String (Transverse)	$v = \sqrt{(T / \mu)}$	Tension T (N)	Linear density μ (kg/m)
Bulk Fluid (Sound, Longitudinal)	$v = \sqrt{(B / \rho)}$	Bulk Modulus B (Pa)	Volume density ρ (kg/m^3)
Solid Rod (Longitudinal Sound)	$v = \sqrt{(Y / \rho)}$	Young's Modulus Y (Pa)	Volume density ρ (kg/m^3)
Ideal Gas (Acoustic Sound)	$v = \sqrt{(\gamma R T_K / M)}$	Adiabatic bulk modulus $B_{ad} = \gamma P$	Gas density $\rho = P M / R T_K$

Chapter 4: Thermodynamic Derivation of Acoustic Speed in Air

In 1687, Sir Isaac Newton attempted to calculate the speed of sound in air assuming that acoustic compressions and rarefactions occur *isothermally* (constant temperature). Under isothermal conditions, Boyle's law applies: $P\,V$ = constant. Differentiating gives $P\,dV + V\,dP = 0$, yielding an isothermal bulk modulus:

$$\mathrm{B_{isothermal}} = -\mathrm{V}\,\frac{\mathrm{dP}}{\mathrm{dV}} = \mathrm{P} \quad \Rightarrow \quad \mathrm{v_{Newton}} = \sqrt{\frac{\mathrm{P}}{\rho}}$$

At standard temperature and pressure (P = 101.3 kPa, ρ = 1.29 kg/m^3), Newton's formula predicts $v \approx 280$ m/s — an error of roughly 16% below the experimentally measured speed of 331 m/s!

▶ The Laplace Adiabatic Correction

In 1816, the French mathematician Pierre-Simon Laplace resolved the discrepancy. Acoustic sound waves in air oscillate at audible frequencies (20 Hz – 20,000 Hz). The rapid alternating compressions and expansions occur far too quickly for heat conduction to equalize temperatures between adjacent wave regions.

Therefore, sound propagation is strictly **adiabatic** (zero heat transfer, $dQ = 0$). The governing thermodynamic relation for a reversible adiabatic process in an ideal gas is:

$$P\,V^{\gamma} = \text{constant}, \qquad \text{where } \gamma = \frac{C_p}{C_v}$$

Taking the differential: $V^{\gamma}\,dP + \gamma\,P\,V^{\gamma-1}\,dV = 0 \Rightarrow dP/dV = -\gamma\,P/V$. Thus, the **adiabatic bulk modulus** is:

$$B_{\text{adiabatic}} = -V\,\frac{dP}{dV} = \gamma\,P$$

Substituting $B_{adiabatic}$ and the ideal gas equation of state $\rho = P\,M/(R\,T_K)$:

$$v = \sqrt{\frac{B_{\text{adiabatic}}}{\rho}} = \sqrt{\frac{\gamma\,P}{\rho}} = \sqrt{\frac{\gamma\,R\,T_K}{M}}$$

For air (primarily diatomic N_2 and O_2), $\gamma \approx 7/5 = 1.40$, $R = 8.314$ J/(mol·K), and mean molar mass $M \approx 0.02897$ kg/mol. At $T = 273.15$ K (0°C):

$$v(0°\text{C}) = \sqrt{\frac{1.40 \times 8.314 \times 273.15}{0.02897}} = 331.3 \text{ m/s}$$

This agrees with empirical measurements. Expanding around 0°C as a function of Celsius temperature T_C:

$$v(T_C) = (331.3 \text{ m/s})\sqrt{1 + \frac{T_C}{273.15}} \approx 331.3 + 0.606\,T_C \quad \text{(m/s)}$$

Part III: Energy Transport, Power Transmission & Sound Intensity

Chapter 5: Mechanical Energetics of Traveling Waves

As a traveling wave propagates along a string, it conveys mechanical energy along the positive x direction. To quantify this energy flux, consider an infinitesimal segment of string dx with mass $dm = \mu\,dx$. The total mechanical energy in the segment is the sum of its kinetic energy dK and elastic potential energy dU.

5.1 Kinetic Energy Density

The segment oscillates transversely with velocity $v_y = \partial y/\partial t = -\omega A \cos(kx - \omega t)$. Its instantaneous kinetic energy is:

$$dK = \frac{1}{2}\,(dm)\,v_y^{\,2} = \frac{1}{2}\,(\mu\,dx)\,[-\omega\,A\cos(kx - \omega t)]^2 = \frac{1}{2}\,\mu\,\omega^2\,A^2\cos^2(kx - \omega t)\,dx$$

5.2 Potential Energy Density

Elastic potential energy is stored because the string is stretched relative to its equilibrium length dx. The stretched length is $ds = \sqrt{(dx^2 + dy^2)} = dx\,\sqrt{[1 + (\partial y/\partial x)^2]} \approx dx\,[1 + (1/2)(\partial y/\partial x)^2]$. The work done by tension T during this elongation is:

$$dU = T\,(ds - dx) = \frac{1}{2}\,T\left(\frac{\partial y}{\partial x}\right)^2 dx$$

Evaluating $\partial y/\partial x = k\,A\cos(kx - \omega t)$ and noting that $T\,k^2 = (\mu\,v^2)(\,\omega/v\,)^2 = \mu\,\omega^2$:

$$dU = \frac{1}{2}\,\mu\,\omega^2\,A^2\cos^2(kx - \omega t)\,dx$$

Key Physical Insight: In a traveling wave, $dK = dU$ everywhere at all times! Both kinetic energy and potential energy peak simultaneously at the equilibrium crossings ($y = 0$), where slope and velocity are greatest. Both vanish simultaneously at the wave crests and troughs ($y = \pm A$), where $\partial y/\partial x = 0$ and $v_y = 0$.

Total Instantaneous Mechanical Energy Density: $$u(x,\,t) = \frac{dE}{dx} = \frac{dK + dU}{dx} = \mu\,\omega^2\,A^2\cos^2(kx - \omega t)$$

5.3 Transmitted Power

Power is the rate at which energy is transmitted past a given coordinate x. The left segment of the string exerts a transverse force $F_y = -T(\partial y/\partial x)$ on the right segment. The instantaneous power transmitted is the product of force and particle velocity:

$$P(x,\,t) = F_y \cdot v_y = \left[-T\,\frac{\partial y}{\partial x}\right]\left[\frac{\partial y}{\partial t}\right] = \mu\,v\,\omega^2\,A^2\cos^2(kx - \omega t)$$

Since the time average of $\cos^2(\theta)$ over any integer number of cycles is exactly 1/2, the **time-averaged transmitted power** is:

$$P_{avg} = \frac{1}{2}\,\mu\,v\,\omega^2\,A^2 = 2\pi^2\,\mu\,v\,f^2\,A^2$$

Universal Proportionality: The average power conveyed by any mechanical wave is strictly proportional to the *square of its frequency* (f^2) and the *square of its amplitude* (A^2).

Chapter 6: Sound Intensity, Inverse-Square Law & Decibel Scale

6.1 Sound Intensity & Geometric Attenuation

Acoustic Intensity I is defined as the time-averaged acoustic power transported per unit surface area perpendicular to the direction of propagation:

$$I = \frac{P_{avg}}{\text{Area}} \qquad (W/m^2)$$

For an idealized, omnidirectional (isotropic) point sound source radiating uniformly into three-dimensional space without dissipation or reflection, the wavefronts form concentric expanding spheres of surface area $4\pi r^2$. Energy conservation mandates the **Inverse-Square Law**:

$$I(r) = \frac{P_{source}}{4\pi r^2} \quad \Rightarrow \quad \frac{I_2}{I_1} = \left(\frac{r_1}{r_2}\right)^2$$

6.2 The Logarithmic Decibel (dB) Sound Level Scale

The human ear is sensitive across a dynamic intensity range of over twelve orders of magnitude (10^{12}). To handle such immense variation, acousticians express intensity on a logarithmic scale called the **Sound Intensity Level** β, measured in **decibels (dB)**:

$$\beta = 10 \log_{10}\left(\frac{I}{I_0}\right) \quad \text{(dB)}, \quad \text{where } I_0 = 1.00 \times 10^{-12}\ \text{W/m}^2$$

The reference constant I_0 represents the nominal **threshold of human hearing** at 1000 Hz, corresponding to β = 0 dB. The **threshold of pain** occurs at I = 1.0 W/m^2, which corresponds to $\beta = 10 \log_{10}(1 / 10^{-12}) = 120$ dB.

The difference in decibels between two sounds of intensities I_1 and I_2 emitted from a point source at distances r_1 and r_2 is:

$$\Delta\beta = \beta_2 - \beta_1 = 10 \log_{10}\left(\frac{I_2}{I_1}\right) = 10 \log_{10}\left[\left(\frac{r_1}{r_2}\right)^2\right] = 20 \log_{10}\left(\frac{r_1}{r_2}\right)$$

Acoustic Environment / Source	Sound Level β (dB)	Intensity I (W/m^2)	Human Sensation / Safety Limits
Threshold of Hearing	0 dB	1.0×10^{-12}	Barely audible in an anechoic chamber.
Rustling Leaves / Gentle Whisper (1 m)	20 – 30 dB	1.0×10^{-10} – 10^{-9}	Very quiet; typical recording studio.
Quiet Bedroom / Library	40 dB	1.0×10^{-8}	Comfortable ambient background.
Normal Conversational Speech (1 m)	60 – 65 dB	1.0×10^{-6} – 3.2×10^{-6}	Ideal clarity; effortlessly intelligible.
Heavy City Traffic / Vacuum Cleaner	75 – 80 dB	3.2×10^{-5} – 1.0×10^{-4}	Noisy; prolonged exposure causes fatigue.
OSHA 8-Hour Permissible Exposure Limit	85 dB	3.16×10^{-4}	Hearing protection recommended for long workdays.
Power Lawn Mower / Motorcycle (1 m)	95 – 100 dB	3.2×10^{-3} – 1.0×10^{-2}	Damage possible after 15–30 minutes unprotected.
Rock Concert / Loud Car Horn	110 – 115 dB	0.10 – 0.32	Extremely loud; rapid acoustic trauma risk.
Threshold of Pain	120 dB	1.0	Immediate physical pain and temporary hearing loss.
Jet Engine Takeoff (30 m distance)	140 dB	100	Rupture of eardrum; permanent acute deafness.

Part IV: The Doppler Effect in Acoustic Systems

Chapter 7: Kinematic Derivation of the Master Doppler Equation

The **Doppler Effect** is the shift in observed wave frequency caused by the relative motion between the wave source, the observer (detector), and the transmitting medium. Named after the Austrian physicist Christian Doppler (1842), this phenomenon occurs universally across all wave types.

Let v be the speed of sound relative to the quiescent medium, f be the emitted source frequency, $T = 1/f$ be the source period, and $\lambda = v/f$ be the quiescent wavelength.

▶ Step 1: Observer Moving Relative to Quiescent Medium (Source at Rest, $v_s = 0$)

When the source is stationary, wavefronts in the medium are concentric spheres spaced by the unaltered wavelength $\lambda = v / f$.

An observer moving with speed v_o directly **toward** the source intercepts wavefronts with an effective relative speed $v_{rel} = v + v_o$. The perceived frequency f' is the rate at which wavefronts pass the observer:

$$f' = \frac{v_{rel}}{\lambda} = \frac{v + v_o}{v / f} = f\left(\frac{v + v_o}{v}\right) \quad \text{[Observer Moving Toward Source]}$$

If the observer moves directly **away** from the stationary source, the relative speed is $v_{rel} = v - v_o$, yielding:

$$f' = f\left(\frac{v - v_o}{v}\right) \quad \text{[Observer Moving Away from Source]}$$

▶ Step 2: Source Moving Relative to Quiescent Medium (Observer at Rest, $v_o = 0$)

Now suppose the observer is stationary while the source moves with speed $v_s < v$. During each period T, the source emits a wavefront and then travels a distance $d = v_s T$ before emitting the next wavefront.

In the forward direction (ahead of the source), the effective wavelength is physically compressed:

$$\lambda' = \lambda - v_s T = \frac{v}{f} - \frac{v_s}{f} = \frac{v - v_s}{f}$$

Because the stationary medium transmits these compressed waves at speed v, the stationary observer detects a higher frequency:

$$f' = \frac{v}{\lambda'} = f\left(\frac{v}{v - v_s}\right) \quad \text{[Source Moving Toward Observer]}$$

Behind the source, wavefronts are stretched: $\lambda'' = \lambda + v_s T = (v + v_s) / f$, yielding a lower observed frequency:

$$f' = f\left(\frac{v}{v + v_s}\right) \quad \text{[Source Moving Away from Observer]}$$

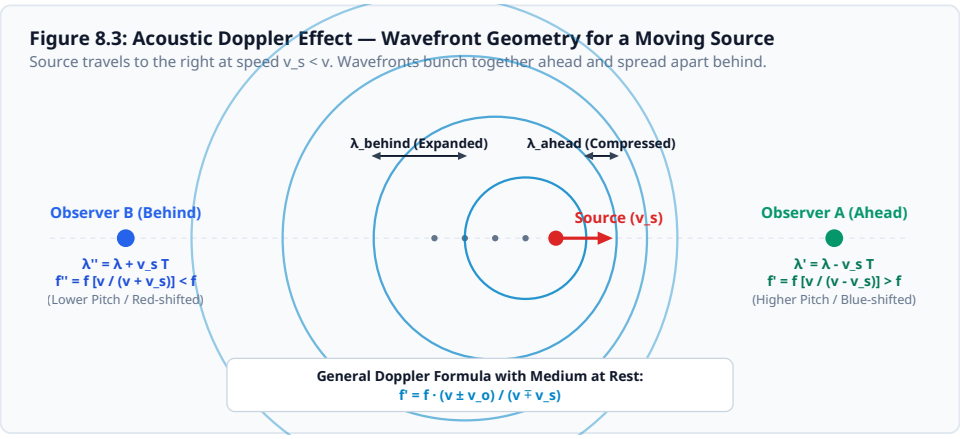


Figure 8.3: Wavefront geometry for a moving sound source ($v_S < v$) illustrating spatial compression ahead and expansion behind.

7.1 The Master Acoustic Doppler Equation

Combining the independent kinematic effects of observer motion and source motion along the line joining them yields the **General Master Doppler Equation**:

$$f' = f \cdot \left(\frac{v \pm v_o}{v \mp v_s} \right)$$

▶ Unambiguous Sign Convention Rule

All velocities v_o and v_s are measured relative to the stationary transmitting medium:

- **Numerator (+ or − v_o):** Choose the **plus sign (+)** if the observer moves *toward* the source (increasing relative encounter rate). Choose the **minus sign (−)** if the observer moves *away* from the source.
- **Denominator (− or + v_s):** Choose the **minus sign (−)** if the source moves *toward* the observer (compressing wavelength). Choose the **plus sign (+)** if the source moves *away* from the observer.

Golden Sanity Check: *Whenever distance between source and observer is decreasing, f' > f (pitch rises). Whenever distance is increasing, f' < f (pitch falls).*

Chapter 8: Medium Motion, Moving Target Echoes & Shock Fronts

8.1 Effect of Moving Medium (Wind Velocity v_W)

If the transmitting medium (air) moves with steady wind velocity v_W along the line connecting source and observer, the effective wave propagation speed through the laboratory frame becomes ($v \pm v_W$):

$$f' = f \cdot \left[\frac{(v + v_W) \pm v_o}{(v + v_W) \mp v_s} \right]$$

8.2 Moving Target Echo Reflection (Sonar & Radar)

When an acoustic pulse of frequency f strikes a moving reflector (e.g., submarine, automobile, or red blood cell in Doppler ultrasound) traveling at speed u toward a stationary transceiver:

1. The moving target acts first as a **moving observer**, detecting frequency $f' = f\,(v + u) / v$.
2. The target reradiates (reflects) this frequency as a **moving source**, so the stationary transceiver receives:

$$f'' = f' \left(\frac{v}{v - u} \right) = f \left(\frac{v + u}{v - u} \right)$$

For low target speeds $u \ll v$, a Taylor series expansion gives the beat frequency Δf between transmitted and reflected signals:

$$\Delta f = f'' - f = f \left[\frac{1 + u/v}{1 - u/v} - 1 \right] \approx f \left[(1 + u/v)(1 + u/v) - 1 \right] \approx 2 f \left(\frac{u}{v} \right)$$

8.3 Supersonic Motion & The Mach Cone

When the source speed equals or exceeds the speed of sound ($v_s \geq v$), wavefronts can no longer outrun the source. Instead, the spherical wavefronts overlap constructingly along a conical envelope known as the **Mach Cone**, forming a high-amplitude **shock wave**:

$$\sin \theta_{\text{Mach}} = \frac{v\,t}{v_s\,t} = \frac{v}{v_s} = \frac{1}{M} \qquad \text{(where } M = v_s / v \text{ is the Mach number)}$$

Part V: Superposition, Boundary Reflections & Standing Resonances

Chapter 9: Boundary Conditions & Standing Wave Derivation

9.1 Reflection Phase Shifts at Physical Boundaries

When a traveling wave pulse encounters a boundary terminating the medium, boundary conditions dictate the amplitude and phase of the reflected wave:

- **Fixed End (Rigid Support):** The boundary is rigidly clamped, enforcing zero displacement: $y(L, t) = 0$. By Newton's Third Law, the support exerts an equal and opposite reaction force on the string, generating an **inverted pulse**. This corresponds to an abrupt **phase inversion of π rad (180°)**. The boundary is an absolute **displacement node**.
- **Free End (Frictionless Ring on Rod):** The string end is completely free to slide transversely without friction. The transverse restoring force must vanish at the boundary: $\partial y/\partial x|_{\text{boundary}} = 0$. The pulse reflects **erect (upright)** with **zero phase shift (0°)**, constructively doubling the boundary displacement. The free end is a **displacement antinode**.

9.2 Mathematical Derivation of a Standing Wave

Consider two harmonic waves of identical amplitude A, frequency ω, and wave number k traveling in opposite directions along a string:

$$y_1(x, t) = A \sin(kx - \omega t) \quad \text{(rightward)}, \qquad y_2(x, t) = A \sin(kx + \omega t) \quad \text{(leftward)}$$

By the **Principle of Linear Superposition**, the resultant wave is the algebraic sum $y(x, t) = y_1(x, t) + y_2(x, t)$. Using the trigonometric identity $\sin \alpha + \sin \beta = 2 \sin[(\alpha + \beta)/2] \cos[(\alpha - \beta)/2]$:

$$y(x, t) = \left[2A \sin(kx) \right] \cos(\omega t)$$

Physical Anatomy of a Standing Wave:

1. **Separation of Variables:** The spatial dependence $[2A \sin(kx)]$ is completely decoupled from the temporal oscillation $\cos(\omega t)$.
2. **Zero Net Energy Transport:** Because energy travels equally in both directions, net energy flux past any point is

exactly zero. Energy remains trapped, sloshing continuously between kinetic and elastic potential forms within each resonant loop.

3. **Nodes (N):** Points of permanent zero motion where $2A \sin(kx) = 0 \Rightarrow kx = n\pi \Rightarrow$

$$x_{\text{Node}} = n \frac{\lambda}{2} \qquad (n = 0, 1, 2, 3, \ldots)$$

Adjacent nodes are spaced by exactly **λ / 2**.

4. **Antinodes (A):** Points of maximum oscillation amplitude ($2A$) where $|\sin(kx)| = 1 \Rightarrow kx = (n + 1/2)\pi \Rightarrow$

$$x_{\text{Antinode}} = \left(n + \frac{1}{2}\right) \frac{\lambda}{2} = (2n + 1) \frac{\lambda}{4} \qquad (n = 0, 1, 2, 3, \ldots)$$

The distance between any node and the nearest antinode is exactly **λ / 4**.

Chapter 10: Resonant Modes in Strings & Organ Pipes

10.1 Stretched String Fixed at Both Ends (Length L)

Because both ends $x = 0$ and $x = L$ are rigidly clamped, they must be nodes: $y(0, t) = 0$ and $y(L, t) = 0$. The condition $2A \sin(kL) = 0$ requires $kL = n\pi$ for $n = 1, 2, 3, \ldots$ Substituting $k = 2\pi / \lambda_n$:

$$\lambda_n = \frac{2L}{n}, \qquad f_n = \frac{v}{\lambda_n} = n \left(\frac{v}{2L}\right) = n f_1 \qquad (n = 1, 2, 3, \ldots)$$

The fundamental frequency (first harmonic, $n = 1$) is $f_1 = v / (2L) = (1/2L)\sqrt{(T / \mu)}$. A string fixed at both ends produces **all integer harmonics** (f_1, $2f_1$, $3f_1$, $4f_1$, …).

10.2 Displacement Wave vs. Pressure Wave in Acoustic Columns

In an acoustic air column, sound waves may be described either as **longitudinal displacement waves** $s(x, t)$ or as **acoustic pressure variations** $\Delta P(x, t)$. The relationship between pressure fluctuation and displacement gradient is given by bulk elasticity: $\Delta P = -B(\partial s / \partial x)$. Consequently:

- At a **closed tube end**: Air molecules cannot move longitudinally ⇒ **Displacement Node**. Pressure fluctuations reach maximum amplitude ⇒ **Pressure Antinode**.
- At an **open tube end**: The tube connects directly to the vast outside atmosphere where pressure remains clamped at P_{atm} ($\Delta P = 0$) ⇒ **Pressure Node**. Air molecules surge freely in and out ⇒ **Displacement Antinode**.

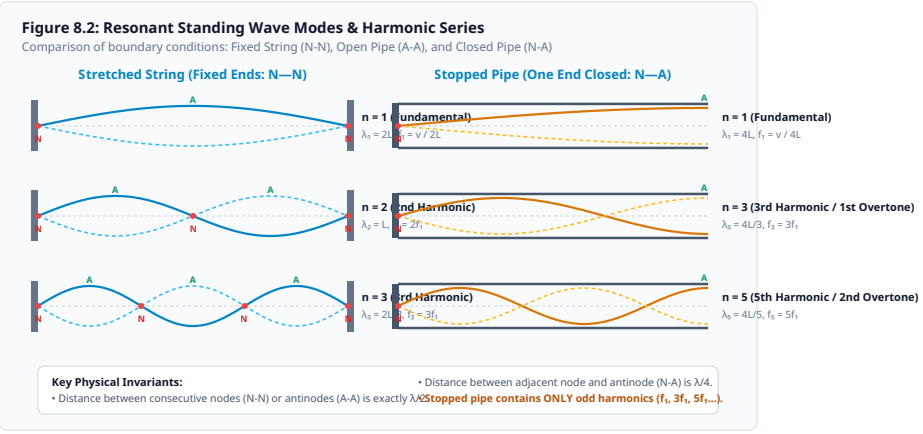


Figure 8.2: Resonant harmonic standing wave modes comparing a string fixed at both ends (Nodes at x = 0, L) against a stopped organ pipe (Node at closed end, Antinode at open end).

10.3 Open-Open Organ Pipe (Open at Both Ends)

Both ends are open to the atmosphere, enforcing displacement antinodes at $x = 0$ and $x = L$. The distance between consecutive antinodes is λ/2, so an integer number of half-wavelengths must fit inside length L:

$$L = n\,\frac{\lambda_n}{2} \quad \Rightarrow \quad \lambda_n = \frac{2L}{n}, \qquad f_n = n\,\frac{v}{2L} = n\,f_1 \qquad (n = 1, 2, 3, \ldots)$$

An open-open organ pipe contains **all integer harmonics** (1st, 2nd, 3rd, 4th, …).

10.4 Open-Closed Organ Pipe (Stopped Pipe: One End Closed, One End Open)

The closed end ($x = 0$) is a displacement node, while the open end ($x = L$) is a displacement antinode. The distance from a node to an antinode is an odd multiple of quarter-wavelengths (λ/4):

$$L = n\,\frac{\lambda_n}{4} \qquad (n = 1, 3, 5, 7, \ldots) \quad \Rightarrow \quad \lambda_n = \frac{4L}{n}, \qquad f_n = n\,\frac{v}{4L} = n\,f_1$$

▶ **Cardinal Acoustic Asymmetry**

- **Fundamental Frequency:** $f_1 = v / (4L)$. This is exactly **half the fundamental frequency** of an open-open pipe of identical physical length. A stopped pipe sounds an octave lower!
- **Harmonic Content:** A stopped pipe possesses **ONLY ODD HARMONICS** (f_1, $3f_1$, $5f_1$, $7f_1$, …). All even harmonics ($2f_1$, $4f_1$, …) are physically impossible because they would require either nodes at both ends or antinodes at both ends. This gives stopped organ pipes and clarinets their characteristically warm, hollow timbre.

Part VI: Temporal Interference & Acoustic Beats

Chapter 11: Modulation Envelopes & The Beat Frequency Equation

When two sound waves of equal amplitude A but slightly different frequencies f_1 and f_2 ($\omega_1 \neq \omega_2$) arrive simultaneously at a listener's ear, they interfere temporally. The resulting acoustic signal alternates periodically between loud (constructive) and soft (destructive) sound. This phenomenon is called **acoustic beats**.

▶ Mathematical Derivation of Beats

Let the two wave displacements at the detector (at $x = 0$) be:

$$y_1(t) = A\cos(\omega_1 t), \qquad y_2(t) = A\cos(\omega_2 t)$$

Applying the prosthaphaeresis trigonometric identity $\cos\alpha + \cos\beta = 2\cos[(\alpha-\beta)/2]\cos[(\alpha+\beta)/2]$:

$$y(t) = y_1(t) + y_2(t) = \left[\, 2A\cos\left(\frac{\omega_1 - \omega_2}{2}\, t\right)\right]\cos\left(\frac{\omega_1 + \omega_2}{2}\, t\right)$$

Defining the **carrier average frequency** $\omega_{avg} = (\omega_1 + \omega_2)/2$ and the **modulation frequency** $\omega_{mod} = |\omega_1 - \omega_2|/2$:

$$y(t) = A_{env}(t)\cos(\omega_{avg} t), \qquad \text{where } A_{env}(t) = 2A\cos\left(\pi\,|f_1 - f_2|\, t\right)$$

The rapid carrier wave vibrates at the average pitch $f_{avg} = (f_1 + f_2)/2$. However, because acoustic sound intensity and perceptual loudness are proportional to the *square of the amplitude envelope* ($I \propto A_{env}^2$), a maximum in sound intensity occurs whenever A_{env} reaches either $+2A$ OR $-2A$!

Since the cosine function reaches ±1 **twice per modulation cycle**, the frequency of loudness peaks — the **Beat Frequency** — is exactly twice the modulation frequency:

$$f_{beat} = 2 \times \frac{|f_1 - f_2|}{2} = |f_1 - f_2|$$

Figure 8.4: Acoustic Beats & Amplitude Modulation Envelope
Superposition of two pure tones (f_1 = 11 Hz, f_2 = 9 Hz) producing beat frequency f_beat = |f_1 - f_2| = 2 Hz

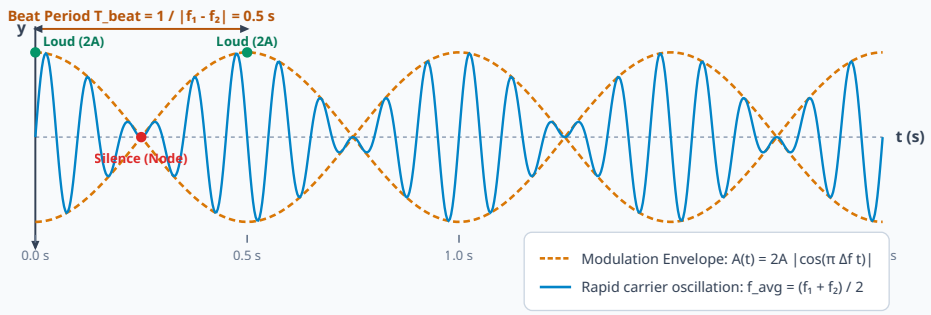


Figure 8.4: Acoustic beat waveform displaying rapid carrier oscillations (f_{avg} = 10 Hz) modulated by a slow envelope producing beat frequency $f_{beat} = |11 - 9| = 2$ Hz.

Part VII: ⚡ Hand Calculation Fast-Tracks

TRICK W1 **The ±3 dB Sound Power & Intensity Doubling Shortcut** **2 Seconds**

Standard Method: Evaluating $\beta_2 - \beta_1 = 10 \log_{10}(2I / I) = 10 \log_{10}(2) = 10 \times 0.30103 = 3.01$ dB.

⚡ **Fast Hand Shortcut:** Memorize that **doubling sound intensity or acoustic power ALWAYS adds exactly +3 dB**. Halving intensity subtracts 3 dB.

Example: If one trumpet produces 75 dB, two identical trumpets produce 75 + 3 = **78 dB** (NOT 150 dB!). Four trumpets produce 78 + 3 = **81 dB**. Eight trumpets produce **84 dB**.

TRICK W2 **The ±10 dB Decade Rule (Perceptual Loudness Doubling)** **Mental Math**

Rule: Every **factor of 10 increase in physical intensity adds exactly +10 dB**.

• 10× intensity = +10 dB • 100× intensity = +20 dB • 1000× intensity = +30 dB • 1,000,000× = +60 dB.

Psychoacoustic Correlate: In human audition, a +10 dB increase is subjectively perceived as *twice as loud*! Hence, a sound 10 times more powerful is perceived as only 2 times louder.

TRICK W3 **Inverse-Square Distance Attenuation (−6 dB per Doubling)** **Instant Ratio**

Rule: In open spherical radiation, doubling distance from a point source ($r \rightarrow 2r$) drops intensity by $(1/2)^2 = 1/4$.

By Trick W1, a quartering of intensity (two successive halvings) decreases the sound level by exactly **6 dB** ($20 \log_{10}(2) \approx 6.02$ dB).

Example: If a loudspeaker produces 90 dB at 10 m, at 20 m it produces 90 − 6 = **84 dB**; at 40 m it produces 84 − 6 = **78 dB**; at 80 m it produces 78 − 6 = **72 dB**.

TRICK W4 **Doppler Direction Sign Mnemonic: “Towards ↑, Away ↓”** **Error-Proof**

Never fumble Doppler plus/minus signs again:

1. **Observer numerator:** Moving *toward* the source must INCREASE the pitch ⇒ **use (+)**. Moving *away* must DECREASE pitch ⇒ **use (−)**.
2. **Source denominator:** Moving *toward* observer must INCREASE the fraction ⇒ make denominator smaller ⇒ **use (−)**. Moving *away* must DECREASE the fraction ⇒ make denominator larger ⇒ **use (+)**.

TRICK W5 **String Tension vs. Pitch Scaling: The Half-Power Law** 3 Seconds

Since $f \propto v = \sqrt{(T / \mu)}$, frequency scales with the square root of tension: $\mathbf{f \propto T^{1/2}}$.

• To raise pitch by one octave (2× frequency), tension must increase by $2^2 = \mathbf{4\times}$.

• For small fractional adjustments: $df / f = (1/2)\, dT / T$. A **2% increase in string tension produces a 1% increase in pitch**.

TRICK W6 **The Organ Pipe Octave Duality (Open vs. Stopped)** Concept Card

For pipes of identical physical length L:

$\mathbf{f_{1,\ open} = 2 \times f_{1,\ stopped}}$ [Because $f_{1,\ open} = v / 2L$, while $f_{1,\ stopped} = v / 4L$].

Practical Rule: Closing one end of an open pipe immediately drops its fundamental pitch by exactly **one octave** and silences all even harmonics!

TRICK W7 **Rapid Acoustic Speed in Air Calculation** Mental Math

Instead of evaluating $\sqrt{(\gamma R T_K / M)}$, use the linear formula:

$$\mathbf{v \approx 331.3 + 0.6\, T_C \quad (m/s)}$$

• At 0°C: $v = 331$ m/s • At 20°C (Room Temp): $v = 331 + 12 = \mathbf{343\ m/s}$ • At 30°C (Hot Summer): $v = 331 + 18 = \mathbf{349\ m/s}$.

TRICK W8 **Beat Tuning Rule: Tracking String Slope** Instrument Tuning

When tuning an instrument string to a reference tuning fork of frequency f_{ref}:

If you **tighten the peg** (which strictly increases f_{string}) and the beat frequency f_{beat} **decreases**, your string was originally *flat* ($f_{string} < f_{ref}$) and is approaching perfect unison.

If tightening causes f_{beat} to **increase**, your string was already *sharp* ($f_{string} > f_{ref}$) and you are driving it further out of tune!

⚡ High-Yield Acoustic Exam Strategy & Pitfalls Checklist

- **Intensity vs. Pressure Decibels:** Decibels for acoustic intensity and power use $\beta = 10 \log_{10}(I/I_0)$, but decibels expressed in acoustic pressure amplitudes use $\beta = 20 \log_{10}(\Delta p / \Delta p_0)$ because intensity scales with pressure squared ($I \propto \Delta p^2$).
- **Boundary Reflection Phase Inversion:** A wave reflecting from a *fixed string clamp* or a *closed rigid pipe end* suffers an instantaneous 180° (π rad) phase inversion. A wave reflecting from a *free ring* or an *open tube end* reflects with **zero phase shift**.
- **Harmonic vs. Overtone Numbering:** For symmetric systems (open-open pipes, fixed-fixed strings), the m-th overtone is the $(m+1)$-th harmonic. For asymmetric systems (stopped organ pipes), only odd harmonics exist: the **1st overtone is the 3rd harmonic ($n = 3$)**, and the 2nd overtone is the 5th harmonic ($n = 5$).
- **Doppler Effect in a Moving Medium:** When an ambient wind of velocity w blows along the line connecting source and observer, the quiescent sound speed v is replaced everywhere by the effective relative speed $v_{eff} = v \pm w$ before applying observer and source kinematic shifts.

Part VIII: Comprehensive Step-by-Step Worked Problems

Worked Problem 8.1: Piano Wire Mechanics, Kinematics & Power Transmission

1. PROBLEM STATEMENT

A high-tensile steel concert piano wire has length L = 1.20 m and total mass m = 18.0 g. It is mounted under an operating tension T = 1200.0 N. A technician plucks the wire, establishing a sinusoidal traveling wave with amplitude A = 2.50 mm and fundamental pitch frequency f = 220.0 Hz (corresponding to musical note A_3).

Calculate: (a) The linear mass density μ and transverse wave velocity v; (b) The wavelength λ and angular wave number k; (c) The maximum transverse particle velocity $v_{y,\,max}$ and compare it to wave speed v; (d) The maximum transverse acceleration $a_{y,\,max}$; (e) The time-averaged power P_{avg} transmitted by the wave.

2. GIVEN PARAMETERS & KNOWNS

- Length L = 1.20 m; Mass m = 18.0 g = 1.80×10^{-2} kg
- Operating Tension T = 1200.0 N = 1.200×10^{3} N
- Wave Amplitude A = 2.50 mm = 2.50×10^{-3} m
- Frequency f = 220.0 Hz; Angular Frequency $\omega = 2\pi f = 2\pi(220.0) = 1382.30$ rad/s

3. SYMBOLIC ALGEBRAIC DERIVATION

(a) Linear mass density: $\mu = m / L$. Wave velocity: $v = \sqrt{(T / \mu)}$.

(b) Wavelength: $\lambda = v / f$. Wave number: $k = 2\pi / \lambda = \omega / v$.

(c) Transverse particle velocity: $v_y = \partial y/\partial t = -\omega A \cos(kx - \omega t) \Rightarrow v_{y,\,max} = \omega A$. Speed ratio: $\eta = v_{y,\,max} / v$.

(d) Transverse acceleration: $a_{y,\,max} = \omega^2 A$.

(e) Time-averaged power: $P_{avg} = (1/2)\,\mu\, v\, \omega^2 A^2$.

4. NUMERICAL CALCULATION WITH EXPLICIT SI UNITS

(a) $\mu = \dfrac{0.0180\ \text{kg}}{1.20\ \text{m}}$ = **0.0150 kg/m** (15.0 g/m).

$v = \sqrt{\dfrac{1200.0\ \text{N}}{0.0150\ \text{kg/m}}}$ = √80,000 = **282.84 m/s**.

(b) $\lambda = \dfrac{282.843\ \text{m/s}}{220.0\ \text{s}^{-1}}$ = **1.2856 m** (128.56 cm).

$k = \dfrac{2\pi}{1.2856\ \text{m}}$ = **4.887 rad/m**.

(c) $v_{y,\,max} = \omega A = (1382.30\ \text{rad/s})(2.50 \times 10^{-3}\ \text{m})$ = **3.456 m/s**.

Ratio $\eta = \dfrac{3.456\ \text{m/s}}{282.84\ \text{m/s}}$ = **0.0122 (1.22%)**.

(d) $a_{y,\,max} = \omega^2 A = (1382.30\ \text{rad/s})^2(2.50 \times 10^{-3}\ \text{m})$ = **4776.9 m/s² ≈ 487 g**.

(e) $P_{avg} = \dfrac{1}{2}(0.0150\ \text{kg/m})(282.843\ \text{m/s})(1382.30\ \text{rad/s})^2(2.50 \times 10^{-3}\ \text{m})^2$ = **25.33 W**.

✓ **Physical Sanity & Boundary Check:** The ratio of maximum particle velocity to wave velocity is $v_{y,\,max} / v$ = 1.22% ≪ 1. This rigorously validates the small-slope, small-amplitude linear wave approximation ($|\partial y/\partial x| = k A = 4.887 \times 0.0025 = 0.0122 \ll 1$) upon which the classical 1D wave equation was derived. The massive acceleration (487 g) is physically standard for tightly strung concert piano strings.

Worked Problem 8.2: Acoustic Radiation, Inverse-Square Decibel Attenuation & Safety Radius

1. PROBLEM STATEMENT

An outdoor concert sound reinforcement horn acts as an omnidirectional point source, radiating an acoustic power P_{source} = 40.0 W at frequency f = 1000 Hz into open spherical space.

Determine: (a) Sound intensity I_1 and sound level β_1 at front-row distance r_1 = 10.0 m; (b) Sound intensity I_2 and sound level β_2 at back-row distance r_2 = 80.0 m; (c) Verify $\Delta\beta$ using the Fast-Track distance doubling rule; (d) Calculate the minimum safety distance r_{safe} beyond which sound level drops below the OSHA 8-hour continuous exposure limit of 85.0 dB.

2. GIVEN PARAMETERS & KNOWNS

- Acoustic Source Power P_{source} = 40.0 W
- Radii: r_1 = 10.0 m, r_2 = 80.0 m
- Reference Hearing Threshold: $I_0 = 1.00 \times 10^{-12}$ W/m^2
- OSHA Safety Threshold: β_{safe} = 85.0 dB

3. SYMBOLIC ALGEBRAIC DERIVATION

(a) Spherical intensity: $I_1 = P_{source} / (4\pi r_1^2)$. Sound level: $\beta_1 = 10 \log_{10}(I_1 / I_0)$.

(b) Intensity: $I_2 = P_{source} / (4\pi r_2^2) = I_1 (r_1 / r_2)^2$. Sound level: $\beta_2 = 10 \log_{10}(I_2 / I_0)$.

(c) Decibel drop: $\Delta\beta = 20 \log_{10}(r_1 / r_2)$.

(d) Inverting decibel formula: $I_{safe} = I_0 \cdot 10^{\beta_{safe} / 10}$.

Since $I_{safe} = P_{source} / (4\pi r_{safe}^2) \Rightarrow r_{safe} = \sqrt{[P_{source} / (4\pi I_{safe})]}$.

4. NUMERICAL CALCULATION WITH EXPLICIT SI UNITS

(a) $I_1 = \dfrac{40.0 \text{ W}}{4\pi (10.0 \text{ m})^2} = \dfrac{40.0}{1256.64}$ = **3.1831×10^{-2} W/m^2** (31.83 mW/m^2).

$\beta_1 = 10 \log_{10}\left(\dfrac{3.1831 \times 10^{-2}}{1.00 \times 10^{-12}}\right) = 10 \log_{10}(3.1831 \times 10^{10}) = 10(10.5028)$ = **105.03 dB**.

(b) Distance ratio $r_2 / r_1 = 80.0 / 10.0 = 8.00$.

$I_2 = I_1 \left(\dfrac{10.0}{80.0}\right)^2 = \dfrac{3.1831 \times 10^{-2}}{64}$ = **4.9736×10^{-4} W/m^2**.

$\beta_2 = 10 \log_{10}\left(\dfrac{4.9736 \times 10^{-4}}{1.00 \times 10^{-12}}\right) = 10 \log_{10}(4.9736 \times 10^{8}) = 10(8.6967)$ = **86.97 dB**.

(c) Fast-Track Verification: Distance increases by 8× = 2^3 (exactly 3 successive doublings of distance). By Trick W3, each doubling subtracts 6.02 dB:

$\Delta\beta = -3 \times 6.0206$ dB = **−18.06 dB**.

β_2 = 105.03 dB − 18.06 dB = **86.97 dB** — an exact match!

(d) Safety Radius:

$I_{safe} = (1.00 \times 10^{-12}) \cdot 10^{85.0 / 10} = 10^{-12} \times 10^{8.5} = 10^{-3.5}$ = **3.1623×10^{-4} W/m^2**.

$r_{safe} = \sqrt{\dfrac{40.0 \text{ W}}{4\pi (3.1623 \times 10^{-4} \text{ W/m}^2)}} = \sqrt{\dfrac{40.0}{3.9738 \times 10^{-3}}} = \sqrt{10065.8}$ = **100.33 m**.

✓ **Physical Sanity & Boundary Check:** At r_2 = 80.0 m, the sound level is 86.97 dB, which is slightly above the 85.0 dB safety limit. Therefore, the safe perimeter must lie slightly beyond 80 m. Our calculated r_{safe} = 100.3 m is physically sound and directly consistent with the logarithmic attenuation profile.

Worked Problem 8.3: Multi-Vehicle Acoustic Doppler Effect & Radar Beat Shift

1. PROBLEM STATEMENT

An emergency police cruiser travels eastward along a straight highway at speed v_s = 35.0 m/s (approx. 126 km/h) with its siren wailing at an intrinsic frequency f_0 = 900.0 Hz. A civilian driver travels westward directly toward the cruiser at speed v_o = 25.0 m/s (90 km/h). The ambient air temperature is T_C = 25.0°C with zero wind.

Calculate: (a) The speed of sound in air v at 25.0°C; (b) The apparent siren pitch $f'_{approach}$ heard by the civilian driver as the vehicles head-on approach each other; (c) The apparent pitch f'_{recede} heard after the two vehicles cross and move away from each other; (d) The net pitch drop $\Delta f'$ heard during the encounter; (e) If the police car operates a forward-facing Doppler speed gun at f_0 = 900.0 Hz, find the beat frequency Δf_{radar} between emitted and echo waves.

2. GIVEN PARAMETERS & KNOWNS

- Source speed: v_s = 35.0 m/s; Observer speed: v_o = 25.0 m/s
- Source Rest Frequency: f_0 = 900.0 Hz
- Ambient Air Temperature: T_C = 25.0°C ⇒ T_K = 298.15 K
- Air constants: γ = 1.40, R = 8.314 J/(mol·K), M = 0.02897 kg/mol

3. SYMBOLIC ALGEBRAIC DERIVATION

(a) Thermodynamic sound speed: $v = \sqrt{(\gamma R T_K / M)} = 331.3\sqrt{(1 + T_C/273.15)}$.

(b) Mutual Approach: $f'_{app} = f_0\left(\frac{v + v_o}{v - v_s}\right)$

(c) Mutual Recession: $f'_{rec} = f_0\left(\frac{v - v_o}{v + v_s}\right)$

(d) Total pitch shift: $\Delta f' = f'_{app} - f'_{rec}$. (e) Reflected echo beat: $f_{echo} = f_0(v + v_o)/(v - v_o)$, $\Delta f_{radar} = f_{echo} - f_0$.

4. NUMERICAL CALCULATION WITH EXPLICIT SI UNITS

(a) v = 331.3 √(1 + 25.0/273.15) = 331.3 √1.0915 = 331.3 × 1.04476 = **346.13 m/s** (Linear approx: 331.3 + 0.606(25) = 346.45 m/s).

(b) Approach: $f'_{app} = 900.0\left(\frac{346.13 + 25.0}{346.13 - 35.0}\right) = 900.0\left(\frac{371.13}{311.13}\right)$ = 900.0(1.19285) = **1073.56 Hz**.

(c) Recession: $f'_{rec} = 900.0\left(\frac{346.13 - 25.0}{346.13 + 35.0}\right) = 900.0\left(\frac{321.13}{381.13}\right)$ = 900.0(0.84257) = **758.32 Hz**.

(d) Net Pitch Drop: Δf' = 1073.56 Hz − 758.32 Hz = **315.24 Hz**.

(e) Radar Beat: $f_{echo} = 900.0\left(\frac{371.13}{321.13}\right)$ = 1040.13 Hz ⇒ Δf_{radar} = 1040.13 − 900.00 = **140.13 Hz** (1st-order: $2f_0v_o/v$ = 130.01 Hz).

✓ **Physical Sanity & Boundary Check:** As the vehicles approach, the heard frequency rises by +173.56 Hz (+19.3%). As they separate, it drops by −141.68 Hz (−15.7%). Notice the asymmetry: approaching frequency shift is always larger in magnitude than receding shift for equal source speeds because the approaching denominator ($v - v_s$) diminishes faster than the receding denominator ($v + v_s$) grows. The total pitch swing of 315 Hz represents a profound drop of nearly 6 semitones (more than a musical tritone), perfectly capturing the everyday acoustic experience of passing emergency sirens.

Worked Problem 8.4: Organ Pipe Resonance, Thermal Pitch Shift & Acoustic Beat Diagnostics

1. PROBLEM STATEMENT

An organ pipe builder crafts a stopped wooden diapason pipe of acoustic length L = 0.850 m (closed at one end, open at the other).

Determine: (a) The fundamental frequency $f_{1,\text{ cold}}$ and the first two overtone frequencies ($f_{3,\text{ cold}}$, $f_{5,\text{ cold}}$) when voiced in an unheated church at T_1 = 0.0°C (v_1 = 331.3 m/s); (b) The new fundamental frequency $f_{1,\text{ warm}}$ when the hall warms to T_2 = 25.0°C (v_2 = 346.13 m/s); (c) The beat frequency f_{beat} heard when this warm pipe is sounded simultaneously with a temperature-invariant precision quartz tuning fork calibrated to f_{fork} = 100.0 Hz.

2. GIVEN PARAMETERS & KNOWNS

- Length of Stopped Pipe: L = 0.850 m (Node at closed end, Antinode at open end)
- Cold Temperature: T_1 = 0.0°C ⇒ Sound velocity v_1 = 331.3 m/s
- Warm Temperature: T_2 = 25.0°C ⇒ Sound velocity v_2 = 346.13 m/s
- Reference Tuning Fork Frequency: f_{fork} = 100.0 Hz

3. SYMBOLIC ALGEBRAIC DERIVATION

(a) For a stopped pipe, resonant boundary conditions require $L = n(\lambda_n/4)$ for odd integers n = 1, 3, 5, …

$$f_n = n\,\frac{v}{4L} = n\,f_1 \qquad (n = 1, 3, 5, \ldots)$$

(b) Thermal frequency shift: Since $f_1 \propto v(T)$, $f_{1,\text{ warm}} = v_2 / (4L)$.

(c) Acoustic beat frequency between warm pipe fundamental and reference fork:

$$f_{\text{beat}} = |f_{1,\text{ warm}} - f_{\text{fork}}|$$

4. NUMERICAL CALCULATION WITH EXPLICIT SI UNITS

(a) Cold Pipe Resonances (0.0°C):

$$f_{1,\text{ cold}} = \frac{v_1}{4L} = \frac{331.3\text{ m/s}}{4(0.850\text{ m})} = \frac{331.3}{3.400} = \textbf{97.44 Hz}.$$

1st Overtone (3rd Harmonic, n = 3): $f_{3,\text{ cold}}$ = 3 × 97.441 Hz = **292.32 Hz**.

2nd Overtone (5th Harmonic, n = 5): $f_{5,\text{ cold}}$ = 5 × 97.441 Hz = **487.21 Hz**.

Note: Harmonics n = 2 and n = 4 are completely absent.

(b) Warm Pipe Resonances (25.0°C):

$$f_{1,\text{ warm}} = \frac{v_2}{4L} = \frac{346.13\text{ m/s}}{3.400\text{ m}} = \textbf{101.80 Hz}.$$

Net thermal frequency rise: $\Delta f_{\text{thermal}}$ = 101.80 Hz − 97.44 Hz = **+4.36 Hz** (+4.47%).

(c) Acoustic Beat Detection:

f_{beat} = |101.80 Hz − 100.00 Hz| = **1.80 Hz** (approx. 18 beats heard every 10 seconds).

Comparison: In the cold hall, the beat frequency would have been |97.44 − 100.0| = **2.56 Hz**.

✓ **Physical Sanity & Boundary Check:** The fundamental wavelength of the pipe is $\lambda_1 = 4L = 4(0.850\text{ m}) = 3.40\text{ m}$. Warming the air increases molecular thermal speeds, driving sound velocity up by 4.48% ((346.13 − 331.3)/331.3 = 0.0448). Because boundary geometry is fixed by the rigid wooden pipe, resonant frequency must scale in direct 1:1 proportionality with wave speed: 97.44 × 1.0448 = 101.80 Hz. The 1.8 Hz beat frequency is slow and distinct, allowing an organ tuner to easily count the pulses and adjust the tuning stopper.

Part IX: Master Waves & Sound Reference Summary Sheet

Master Formula & Constant Inventory

Physical Quantity / Law	Master Symbolic Equation	Key Variables & Units	Primary Application Context
Harmonic Wave Function	$y(x, t) = A \sin(kx - \omega t + \varphi)$	A (m), k (rad/m), ω (rad/s)	Traveling sinusoidal wave in $+x$ direction.
1D Classical Wave Equation	$\partial^2 y/\partial x^2 = (1/v^2)\, \partial^2 y/\partial t^2$	$v = \omega/k = \lambda f$ (m/s)	Governs non-dispersive small-amplitude 1D wave propagation.
String Wave Speed	$v = \sqrt{(T / \mu)}$	T = tension (N), μ = linear mass (kg/m)	Transverse wave propagation on tensioned strings & piano wires.
Sound Speed in Bulk Fluids	$v = \sqrt{(B / \rho)}$	B = bulk modulus (Pa), ρ = density (kg/m^3)	Acoustic longitudinal waves in liquids & unconfined fluids.
Sound Speed in Ideal Gases	$v = \sqrt{(\gamma R T_K / M)}$	$\gamma = C_p/C_v$, R = 8.314 J/(mol·K), M (kg/mol)	Thermodynamic adiabatic acoustic speed in gases (air γ = 1.40).
Linear Air Speed Approx.	$v(T_C) \approx 331.3 + 0.606\, T_C$	T_C = Celsius temperature (°C)	Rapid hand calculation of acoustic speed in room-temperature air.
Average Transmitted Power	$P_{avg} = (1/2)\, \mu\, v\, \omega^2\, A^2$	P (W), $\omega = 2\pi f$, A = amplitude (m)	Mechanical energy rate transmitted along a harmonic string wave.
Acoustic Intensity Definition	$I = \Delta p_{max}{}^2 / (2\, \rho\, v) = P / \text{Area}$	I (W/m^2), Δp_{max} = pressure amplitude (Pa)	Time-averaged energy flux per unit cross-sectional wavefront area.
Inverse-Square Intensity	$I(r) = P_{source} / (4\pi r^2)$	I (W/m^2), P (W), r = distance (m)	Acoustic power distribution from an isotropic point radiator.
Decibel Sound Level	$\beta = 10 \log_{10}(I / I_0)$	$I_0 = 1.00 \times 10^{-12}$ W/m^2 (0 dB)	Logarithmic human auditory perception scale.
Decibel Distance Drop	$\Delta\beta = 20 \log_{10}(r_1 / r_2)$	r_1, r_2 = radii from point source	Drop in sound level between two radial distances (−6.02 dB per doubling).
General Master Doppler	$f' = f\, [\, (v \pm v_o) / (v \mp v_s)\,]$	v_o = observer speed, v_s = source speed	Kinematic frequency shift (Toward = +, −; Away = −, +).
Radar / Sonar Echo Shift	$f'' = f\, [\, (v + u) / (v - u)\,] \approx f(1 + 2u/v)$	u = approach velocity of target	Two-way reflection Doppler shift for radar speed guns & ultrasound.
Mach Cone Shock Wave	$\sin \theta_{Mach} = v / v_s = 1 / M$	$M = v_s / v \geq 1$ (Mach number)	Semi-angle of conical sonic boom wavefront for supersonic motion.
Standing Wave Superposition	$y(x, t) = [2A \sin(kx)] \cos(\omega t)$	Nodes at $x = n\lambda/2$, Antinodes at $(2n+1)\lambda/4$	Superposition of counter-propagating waves; zero net energy flux.
String Fixed at Both Ends	$f_n = n\, (v / 2L)$, $\lambda_n = 2L/n$	n = 1, 2, 3, 4, … (all integers)	Resonant frequencies on cords with fixed boundary conditions.
Open-Open Organ Pipe	$f_n = n\, (v / 2L)$, $\lambda_n = 2L/n$	n = 1, 2, 3, 4, … (displacement antinodes)	Produces complete harmonic spectrum (fundamental $f_1 = v/2L$).
Stopped Organ Pipe (N–A)	$f_n = n\, (v / 4L)$ (n = 1, 3, 5, …)	Closed end = node; Open end = antinode	ONLY ODD harmonics present; fundamental sounds an octave lower.
Acoustic Beat Frequency	$f_{beat} = \lvert f_1 - f_2 \rvert$	$T_{beat} = 1 / \lvert f_1 - f_2 \rvert$	Modulation envelope pulsation frequency of two interfering tones.

Characteristic Acoustic Media Properties & Decibel Perception Scale

Medium & State	Density ρ (kg/m^3)	Wave Speed v (m/s)	Acoustic Impedance Z (Rayl)	Acoustic Application Context
Air (Dry, 0°C, 101.3 kPa)	1.293	331.3	428.4	Standard STP atmospheric reference state.
Air (Dry, 20°C, 101.3 kPa)	1.204	343.4	413.5	Standard room temperature ambient acoustics.
Helium Gas (0°C, 101.3 kPa)	0.178	965	171.8	Low density, high acoustic velocity (vocal pitch shift).
Pure Distilled Water (20°C)	998	1482	1.48×10^6	Standard unconfined liquid propagation.
Seawater (20°C, 3.5% Salinity)	1025	1522	1.56×10^6	Ocean acoustics, marine SONAR propagation.
Human Soft Tissue (Mean)	1060	1540	1.63×10^6	Diagnostic medical ultrasound imaging standard.
Compact Human Bone	1900	4080	7.75×10^6	High acoustic impedance; strong ultrasonic shadowing.
Rolled Aluminium	2700	5000	1.35×10^7	Lightweight aerospace structural metal.
Structural Carbon Steel	7850	5960	4.68×10^7	Longitudinal acoustic transmission in construction solids.
Pure Lead	11,300	1200	1.36×10^7	Heavy, highly compliant acoustic damping medium.

Boundary Energy Reflection & Transmission: At a normal interface between media of impedances Z_1 and Z_2, the energy reflection coefficient is $R = [(Z_2 - Z_1)/(Z_2 + Z_1)]^2$ and transmission coefficient is $T = 4\,Z_1\,Z_2 / (Z_1 + Z_2)^2$ with $R + T = 1$. Between air (413 Rayl) and water (1.48×10^6 Rayl), R = 99.89%, explaining why ultrasonic imaging strictly requires acoustic impedance-matching gel.

Sound Level β	Intensity I (W/m^2)	RMS Pressure Δp_{rms}	Representative Acoustic Source & Physiological Effect
0 dB	1.00×10^{-12}	20.0 μPa	Absolute threshold of unimpaired human hearing at 1.0 kHz.
20 dB	1.00×10^{-10}	200 μPa	Faint whisper at 1 m distance; rustling leaves in gentle breeze.
40 dB	1.00×10^{-8}	2.00 mPa	Quiet suburban bedroom at night; public library reading room.
60 dB	1.00×10^{-6}	20.0 mPa	Normal conversational speech at 1 m distance.
80 dB	1.00×10^{-4}	0.200 Pa	Heavy urban curb traffic; loud morning alarm clock at 0.5 m.
85 dB	3.16×10^{-4}	0.356 Pa	OSHA 8-hour continuous occupational safety threshold (hearing protection required).
100 dB	1.00×10^{-2}	2.00 Pa	Pneumatic jackhammer at 2 m; passing subway train platform.
120 dB	1.00	20.0 Pa	Front row at heavy rock concert; threshold of auditory discomfort.
130 dB	10.0	63.2 Pa	Industrial air horn at 3 m; acute threshold of physiological ear pain.
140 dB	100	200 Pa	Jet aircraft engine takeoff at 30 m; immediate permanent cochlear trauma.
160 dB	1.00×10^4	2000 Pa	High-caliber military rifle muzzle blast at ear level; instant eardrum rupture.

▶ Fundamental Physical Constants for Acoustic Mechanics

- **Threshold of Human Hearing:** $I_0 = 1.00 \times 10^{-12}$ W/m^2 (0 dB $\Leftrightarrow$ $\Delta p_{0,\,\mathrm{rms}} = 2.00 \times 10^{-5}$ Pa)
- **Threshold of Physical Pain:** $I_{\mathrm{pain}} = 1.00$ W/m^2 (120 dB $\Leftrightarrow$ $\Delta p_{\mathrm{pain,\,rms}} = 20.0$ Pa)
- **Adiabatic Index for Diatomic Air:** $\gamma = C_p / C_v \approx 7/5 = 1.400$
- **Universal Molar Gas Constant:** $R = 8.31446$ J/(mol·K)
- **Mean Molar Mass of Dry Air:** $M_{\mathrm{air}} \approx 0.02897$ kg/mol = 28.97 g/mol
- **Speed of Sound in Dry Air at 0.0°C:** $v_0 = 331.3$ m/s (1193 km/h)
- **Speed of Sound in Pure Water at 20°C:** $v_{\mathrm{water}} \approx 1482$ m/s | **Seawater (20°C):** $v \approx 1522$ m/s
- **Speed of Sound in Structural Steel at 20°C:** $v_{\mathrm{steel}} \approx 5960$ m/s